Calibrated Pseudo-Noisy Supervision for ESPI Denoising with Severity-Controlled Evaluation

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Abstract

Electronic Speckle Pattern Interferometry (ESPI) denoising is constrained by synthetic-to-real mismatch: synthetic corruptions that appear plausible at the image level do not necessarily provide useful supervision for structure-preserving restoration. We propose a calibrated pseudo-noisy supervision framework for ESPI based on a physics-informed corruption cascade comprising multiplicative speckle, Poisson shot noise, and additive Gaussian readout noise. Generator parameters are calibrated from real single-shot / averaged image pairs via moment matching, and the resulting synthetic regimes are evaluated with a fixed lightweight DnCNN-Lite-ECA probe rather than through architecture search. We study a canonical Mid regime, a bounded High-regime extension, and a severity-controlled ablation of the corruption chain. Under severity-matched conditions validated across both a 240-pair random split and an expanded 699-image specimen-held-out protocol, the full cascade is better aligned with the probe's primary validation objective than matched simpler baselines, although the advantage is metric-dependent rather than universal. A bounded wood-to-carbon transfer study further shows that the favorable ranking remains visible under material shift, and paired image-level carbon analysis reinforces this transfer-based reading on the primary metric. We further introduce a reproducibility-oriented v3.2 execution layer that preserves the frozen baseline while enabling deterministic replay, order invariance, and regime-aware calibration. The paper contributes a principled synthetic-supervision framework for ESPI and a fixed-probe methodology for evaluating corruption models under domain mismatch.

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Computer vision*; • **Applied computing** → *Physical sciences and engineering*.

Keywords

ESPI, denoising, synthetic supervision, calibrated noise modeling, domain mismatch, reproducibility

1 Introduction

Electronic Speckle Pattern Interferometry (ESPI) produces fringe-structured images whose useful information is encoded in subtle phase- and amplitude-dependent spatial patterns. In practice, single-shot ESPI acquisitions are strongly affected by speckle texture, detector noise, and acquisition instability. Denoising in this setting is therefore not a generic image-enhancement problem: aggressive smoothing may improve pixel-level similarity while suppressing fringe micro-structure that remains operationally important for downstream interpretation.

A central difficulty is the lack of abundant paired noisy/clean data. While averaged acquisitions can provide lower-noise references, direct supervised training remains limited by dataset size and by the mismatch between real noisy images and naively synthesized corruption models. In preliminary experiments, weakly calibrated synthetic regimes yielded only marginal or unstable practical gains, suggesting that the main bottleneck was not simply denoiser capacity but the fidelity of the synthetic supervision itself. This problem is not unique to ESPI. In many scientific imaging settings, paired noisy/clean data are too scarce or too expensive to support conventional supervised training, making synthetic corruption one of the few practical sources of supervision. In such cases, the quality of the synthetic regime directly affects the quality of the learned restoration, yet the design of the corruption model is rarely treated as a first-class experimental question. To make the point concrete: in the unmatched ablation, a Gaussian-only corruption reaches 22.56 dB generator-side PSNR, apparently far stronger than the full cascade at 17.96 dB. Yet once severity is controlled, the full cascade yields a probe-side validation loss of 0.0850, compared with 0.0866 for the Gaussian-only baseline. This discrepancy, invisible at the generator level, motivates the present study.

Rather than proposing a new restoration backbone, we focus on *synthetic supervision as the object of study*. We formulate a calibrated pseudo-noisy ESPI generator based on a physics-informed corruption cascade; evaluate it through a fixed lightweight probe denoiser; and study both regime sensitivity and corruption-model ablations under controlled severity. This framing shifts the question from "which denoiser architecture is best?" to "what kind of synthetic corruption model yields useful and reproducible supervision under domain mismatch?" In that sense, ESPI functions here as a demanding testbed for a more general AI question: how should synthetic supervision be designed and evaluated when paired real data are scarce, structured noise is dominant, and image-level plausibility alone is not a reliable indicator of downstream learning value? The answer proposed in this paper is to calibrate the corruption model against real acquisitions and to evaluate it through a fixed probe that makes the supervision regime itself experimentally visible.

The central methodological challenge is that image-level plausibility is not enough. A synthetic corruption model may look visually reasonable, or even score highly on generator-side similarity metrics, while still providing supervision that is poorly aligned with the behavior required at test time. This is especially problematic in ESPI, where useful structure is concentrated in fringe morphology and where over-simplified corruption models can alter the balance between global smoothness, local edge fidelity, and speckle-driven texture. For this reason, the present work explicitly separates *generator-side fidelity* from *probe-side utility*: the former measures how a synthetic corruption compares to a low-noise reference, whereas the latter measures whether the resulting supervision actually supports useful denoiser learning.

A second challenge is experimental reproducibility. Synthetic-data pipelines are often treated as lightweight implementation

details, even though their outputs may depend strongly on execution order, hidden randomization policies, or undocumented preprocessing choices. In a setting where the generator itself is the scientific object, such behavior is unacceptable. The proposed framework therefore combines calibrated corruption modeling with a reproducibility-oriented execution layer that preserves the frozen scientific baseline while making regeneration and comparison deterministic and auditable.

The contributions of this paper are fourfold:

(1) We formulate a calibrated pseudo-noisy generator for ESPI using a staged corruption model with multiplicative speckle, Poisson shot noise, and Gaussian readout noise, with parameters estimated from real single-shot / averaged pairs via moment matching.
(2) We introduce a *fixed-probe evaluation paradigm* that separates generator-side fidelity from probe-side utility and study the generator in a canonical Mid regime and a bounded High-regime extension.
(3) We show, through severity-controlled ablation validated at two independent experimental scales and under bounded wood-to-carbon material shift, that the full corruption chain is better aligned with the probe's primary validation objective than matched simpler baselines, although gains remain metric-dependent.
(4) We add a reproducibility-oriented execution layer (v3.2) that preserves the frozen baseline while enabling deterministic replay, order invariance, and regime-aware calibration.

These contributions are methodological rather than architectural: the paper does not propose a new denoising backbone, but instead provides a principled framework for designing, calibrating, and evaluating synthetic supervision under domain mismatch. ESPI serves as a demanding testbed, but the evaluation template—calibration against real pairs, severity-controlled comparison of corruption families, and fixed-probe assessment of downstream utility—is relevant to other data-scarce scientific imaging settings.

2 Related Work

ESPI and related coherent imaging modalities exhibit structure-dependent speckle behavior and strong sensitivity to local fringe morphology [1]. Denoising in interferometric settings must therefore preserve physically meaningful structure rather than optimize only generic image-quality metrics. Classical adaptive filters such as the Lee filter [9] established important baselines for multiplicative-noise suppression. More recently, learning-based methods have extended these approaches with convolutional models designed for severe speckle conditions [10–12], wrapped-phase data [13–15], and dense or uneven fringe patterns [16–18]. These studies show that modern CNNs can improve ESPI restoration, but they largely assume that an appropriate supervision regime already exists.

A second relevant line of work concerns sensor-noise modeling and synthetic supervision. Mixed Poisson–Gaussian models are well established in computational imaging [7, 8], but coherent interferometric data introduce an additional multiplicative component that cannot be reduced to a single additive perturbation. Self-supervised paradigms such as Noise2Noise [5] and Noise2Void [6] have shown that effective restoration can sometimes be learned without fully clean targets, further highlighting that supervision quality is as important as architecture. At the same time, synthetic-to-real transfer work in rendering and simulation has repeatedly shown that visually plausible synthetic data can fail when the underlying nuisance process is poorly matched to the target domain [19, 20]. This observation is directly relevant here: the challenge is not simply to synthesize noisy images, but to synthesize a corruption regime that is useful for downstream denoiser learning.

Residual CNN denoisers such as DnCNN [2] and lightweight channel-attention mechanisms such as ECA [3] provide efficient restoration backbones, but most of this literature assumes that meaningful supervision is already available. In contrast, the present work deliberately fixes the denoiser and studies the supervision regime itself. This design choice prevents architectural tuning from obscuring the effect of the synthetic corruption model and turns the denoiser into an evaluation instrument rather than the main object of optimization. To our knowledge, no prior ESPI denoising study has explicitly separated generator-side fidelity from probe-side utility under severity-controlled comparison, nor validated the corruption-model choice across two independent experimental scales and under bounded wood-to-carbon material shift. This distinction is important because several strong ESPI denoising papers optimize restoration quality with a chosen supervision regime, whereas the present work asks a different question: which supervision regime itself is worth trusting once architecture search is factored out? By making the denoiser fixed and the corruption model variable, the paper changes the unit of analysis from network design to supervision design.

3 Method

3.1 Physics-Informed Corruption Model

Let x denote an averaged ESPI reference image used as a low-noise proxy target. We generate a pseudo-noisy sample y through the staged corruption chain

$$y = G_\sigma\big(P_p\big(S_k\big(B_f(x)\big)\big)\big), \tag{1}$$

where B_f denotes optional frequency-aware blur, S_k multiplicative speckle with contrast parameter k, P_p Poisson shot noise with photon-scale parameter p, and G_σ additive Gaussian readout noise with standard deviation σ. The canonical path used in the main experiments reduces to $x \to x_s \to x_p \to y$, since the blur stage is inactive in the frozen baseline.

This staged formulation is deliberate. A single additive corruption cannot adequately represent the mixed, signal-dependent nuisance process observed in ESPI. The multiplicative term models speckle-driven texture, the Poisson stage approximates detection stochasticity, and the Gaussian stage captures readout noise. Figure 1 should be read top-down and left-to-right: the calibration block sits above the chain because it provides the frozen parameters injected into the downstream corruption stages. In other words, the generator is designed to preserve the qualitative structure of the acquisition process even though it remains an intentionally simplified model.

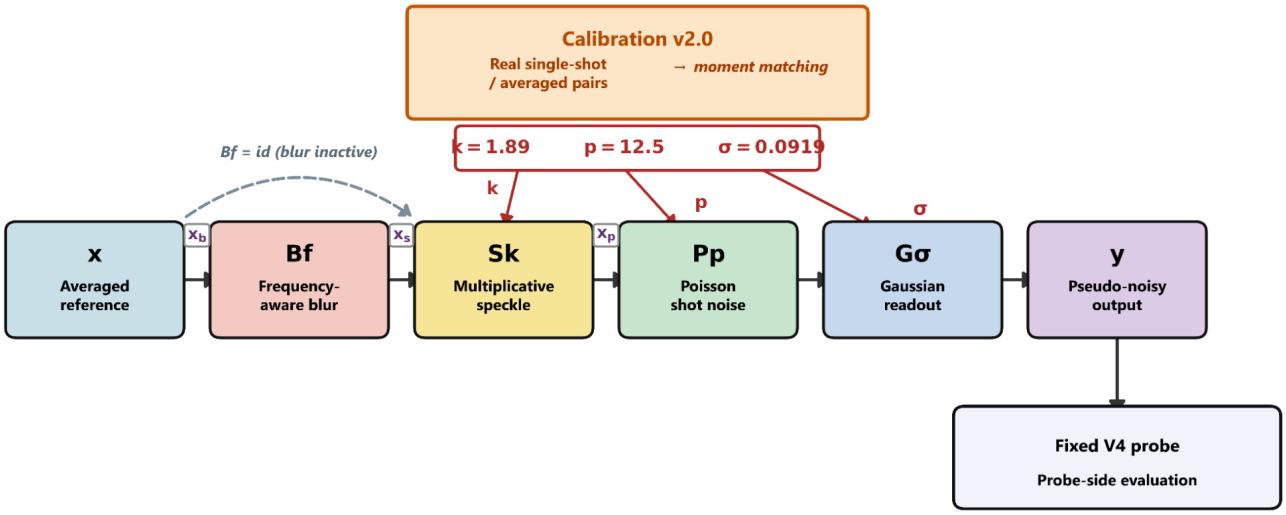


Figure 1: Calibrated pseudo-noisy ESPI generation pipeline. The upper calibration block indicates that the frozen parameters (k, p, σ) are estimated upstream from real single-shot / averaged pairs and then injected into the corruption chain below. An averaged reference is transformed through optional blur, multiplicative speckle, Poisson shot noise, and Gaussian readout, and the resulting synthetic supervision is evaluated by a fixed V4 probe [2–4].

3.2 Calibration from Real Pairs

Generator parameters are not chosen heuristically. Instead, they are calibrated from real single-shot / averaged image pairs by moment matching and summary-statistic alignment. In abstract form, if $\Phi_{k,p,\sigma}$ denotes the corruption pipeline, then calibration seeks parameter values that minimize a discrepancy between synthetic samples generated from averaged references and the corresponding real single-shot observations. The canonical Mid-regime operating point obtained in this way is $k = 1.89$, $p = 12.5$, and $\sigma = 0.0919$; these values define the frozen parameterization used throughout the main study.

Calibration matters because weakly constrained synthetic corruption can look plausible while remaining functionally misaligned with real data. The present framework therefore treats calibration as part of the scientific contribution rather than as a secondary implementation detail. The same logic motivates the later v3.2 execution layer: once the corruption model becomes the object of study, parameter traceability and deterministic replay become essential. In the methodology-upgrade branch, this same calibration logic was extended to global and by-regime modes. The global mode yielded a single parameter block $(k, p, \sigma) = (2.111, 24.765, 0.086)$, whereas the by-regime mode produced distinct Mid and High blocks $(2.015, 15.221, 0.084)$ and $(2.224, 27.184, 0.087)$, respectively. These values did not radically change aggregate image quality, but they demonstrated that regime-aware calibration leaves a real and traceable footprint on the generator rather than acting as a merely symbolic refinement.

3.3 Component-Wise View and Fixed-Probe Evaluation

The generator can be studied either as the full cascade or through controlled component removals. We use four corruption conditions: A1 full cascade, A2 Poisson+Gaussian, A3 Gaussian only, and A4 Speckle+Gaussian. The unmatched comparison asks how these models differ at the generator level. The key scientific comparison, however, is the severity-controlled probe-side ablation: A2 and A3 are re-tuned so that their generator-side PSNR matches A1 within ±0.01 dB, allowing corruption structure to be studied separately from corruption magnitude.

All synthetic regimes are evaluated with a fixed V4 DnCNN-Lite-ECA probe. This is a deliberate methodological choice. The probe is not treated as a new architectural contribution; it functions as a stable instrument for measuring the usefulness of the supervision regime. By fixing the denoiser, we avoid attributing gains from better synthetic supervision to backbone-specific tuning. The fixed-probe design also enforces comparability across canonical, ablated, and regime-aware settings, making the supervision regime rather than the network the main manipulated variable. This design choice is especially important for the severity-controlled experiments. Once A2 and A3 are re-tuned to match the full cascade on generator-side PSNR, the remaining performance differences are no longer dominated by gross corruption magnitude. Instead, they reflect whether the probe benefits from the specific structure of the corruption model, which is exactly the question the paper aims to answer.

3.4 Regimes and Scope of the Comparison

We distinguish between a canonical *Mid* regime and a bounded *High* extension. The Mid regime is the frozen baseline: it has the most controlled calibration path and the clearest probe-side evaluation protocol. The High regime is introduced to test whether the generator remains learnable beyond its canonical region. This comparison is explicitly bounded: informative about operational extension, but not a fully symmetric benchmark. The fixed-probe perspective is important here because it enforces a consistent learning objective across all generator variants.

4 Experimental Setup

4.1 Data and Protocols

We use averaged ESPI images of wood-board specimens as low-noise references, derived from measurements acquired under the laser-based interferometric framework described in Bakarezos et al. [21], and generate pseudo-noisy observations through the calibrated corruption chain. Three complementary settings are considered.

Canonical Mid/High evaluation. The Mid regime uses 240 pseudo-noisy / clean pairs (204 train, 36 validation, seed 42). The High regime uses 113 pairs (97 train, 16 validation). Both use the fixed V4 probe under the same training protocol. The Mid setting is treated as canonical because its calibration and downstream validation path are the best controlled and were frozen earliest in the study. The High setting is retained as a bounded extension rather than a replacement baseline.

Severity-matched ablation. Four generator configurations are compared: A1 full cascade, A2 Poisson+Gaussian, A3 Gaussian only, and A4 Speckle+Gaussian. Because unmatched baselines differ substantially in corruption severity, the probe-side comparison is repeated under severity-controlled matching, where the Gaussian strength in A2 and A3 is increased to match A1's generator-side PSNR within ±0.01 dB. The original random-split study used 240 pairs (204 train, 36 validation); the strengthened pilot uses a specimen-held-out protocol on a larger image pool.

Expanded specimen-held-out protocol. A larger pilot uses 699 averaged images from three wood specimens (W01, W02, W03). W02 and W03 provide 456 training images; W01 provides 243 held-out validation images. The original strengthened pilot uses tile size 256 and seed 42; the later multi-seed extension preserves the same split and training recipe while varying only the training seed. The split is regime-balanced at the subset level, with 265 Mid and 191 High images in train, and 131 Mid and 112 High images in validation. This protocol is scientifically stronger than the original random split because it prevents specimen leakage between train and validation and therefore asks whether the learned restoration behavior survives a small but meaningful domain shift at the specimen level. The strengthened pilot is not a full cross-validation campaign, but it provides a stricter probe-side check than the original 240-pair random split.

4.2 Metrics and Evaluation Logic

We report PSNR, SSIM, and EdgeF1 as image-level restoration metrics, together with validation loss for the fixed probe. Validation loss is treated as the probe's primary optimization-aligned criterion, while EdgeF1 captures fringe-relevant edge preservation. Generator-side metrics diagnose severity; the main scientific comparison is probe-side under severity control.

The evaluation logic follows a strict hierarchy. Generator-side metrics are used to diagnose whether a corruption family is milder or harsher, but they are not treated as sufficient evidence of useful supervision. The main scientific comparison is probe-side: once corruption severity is controlled, does the full calibrated corruption model induce better learning behavior than matched simpler baselines? Validation loss is treated as the probe's primary criterion because it is directly aligned with the learning objective used during training. PSNR gives a conservative global-fidelity view, SSIM captures broader structural similarity, and EdgeF1 is included because fringe-preserving denoising can improve edge retention even when it does not optimize the primary objective. This distinction is the reason why the paper reports both unmatched generator-side results and severity-controlled probe-side results rather than relying on one metric family alone.

4.3 Implementation and Integrity Checks

All probe-side comparisons use the same lightweight V4 DnCNN-Lite-ECA probe and stable data-routing logic; outside the multi-seed extension in Section 5.3, the training seed is fixed at 42. In the expanded specimen-held-out pilot, the generator outputs were required to satisfy strict one-to-one mapping constraints between clean references, manifests, and noisy outputs. This design choice was important because an earlier pilot branch exposed filename-collision failure modes in the validation split. The final manifest-driven build eliminated duplicate noisy paths, ensured exact count agreement between metadata and physical files, and preserved deterministic replay under the v3.2 execution policy. The fixed-probe protocol therefore controls not only the network family but also the optimization context: the same lightweight restoration backbone, training seed, tile size, and split definitions are preserved across corruption conditions wherever the comparison is intended to be fair. This avoids the common failure mode in synthetic-data studies where the corruption model, training recipe, and architecture are all changed simultaneously, making causal attribution impossible. In addition to the training-time validation path used by the pilot wrapper, we later performed a stricter post hoc reevaluation of the selected best checkpoints on the explicit held-out W01 noisy/clean pair manifests. The two views are complementary rather than numerically interchangeable: the former governs checkpoint selection during training, whereas the latter provides a direct held-out pair-based utility check of the same specimen after training.

5 Results

The Mid/High comparison is presented first as contextual evidence; the main scientific comparison is the severity-controlled specimen-held-out study in Section 5.3, which is then extended by bounded carbon-composite material-shift validation in Section 5.4.

Table 1: Compact summary of canonical Mid and bounded High evaluation under the fixed V4 probe.

Regime	Ep.	Loss	PSNR	SSIM	E-F1
Mid ($n = 36$)	14	0.0116	19.58	0.318	0.441
High ($n = 16$)	2	0.0852	24.91	0.787	0.795

5.1 Canonical Mid and High Evaluation

Table 1 summarizes the main probe-side results for the two regimes. To keep the summary compact, validation-set sizes are encoded directly in the regime labels. The High regime remains learnable and does not collapse operationally. However, the Mid–High comparison carries explicit asymmetry caveats: different tile sizes, different validation-set sizes, and different metric provenance. Specifically, the Mid validation metrics were recovered post hoc from the frozen best checkpoint because per-image validation payloads were not preserved directly in the original run logs. The comparison should therefore be interpreted as bounded evidence of regime extension rather than a fully symmetric benchmark.

Diagnostic evidence suggests that the gap is more plausibly associated with structural differences between the regimes than with trivial input-severity differences. The High clean references are modestly smoother and slightly lower in high-frequency content, while the input PSNR is actually lower for High (16.20 vs. 17.82 dB). In that sense, the High result does not indicate an easier corruption problem, but a different operating region with its own structural characteristics. Concretely, the Mid and High diagnostic package showed input SSIM values of 0.526 and 0.353, noise-standard-deviation proxies of 0.129 and 0.151, clean edge densities of 0.158 and 0.142, and clean high-frequency spectral ratios of 0.163 and 0.143, respectively. These numbers support a nuanced reading: High begins from a harsher noisy baseline, yet its clean references are structurally smoother. The observed probe-side gain is therefore more plausibly explained by structural/statistical regime differences than by a trivial reduction in corruption difficulty. The appropriate interpretation is therefore conservative: the generator extends to the High regime with limitations. The result is useful because it shows that the calibrated corruption model does not immediately become degenerate outside its canonical region, yet it does not license a strong claim of regime symmetry.

5.2 Unmatched Generator-Side Ablation

Before severity control, simpler baselines appear stronger on generator-side fidelity because they apply weaker or less structured corruption. Table 2 shows the unmatched generator-side comparison under the canonical parameterization. A3 Gaussian-only yields the highest PSNR, SSIM, and EdgeF1 values, but this does not imply that it provides the best supervision for denoiser training; it mainly reflects a lower effective degradation level.

The corresponding MSE decomposition showed that the multiplicative speckle stage accounts for approximately 58% of total corruption intensity, while the Gaussian component alone contributes less than 9%. This immediately suggests that image-level fidelity alone is a poor criterion for comparing corruption models: a simplified corruption can be closer to the clean reference simply because it is less severe. This unmatched table is therefore intentionally included as a warning against naïve interpretation rather than as evidence of superiority. The A4 Speckle+Gaussian variant is also informative: despite retaining the multiplicative stage, it still appears substantially cleaner than the full cascade, which indicates that the Poisson component is not a cosmetic addition but a material contributor to the overall severity profile. In other words, the full cascade is not "harder by accident"; it is harder because it preserves the composition of nuisance sources that the reduced baselines remove. In particular, it shows why evaluation of synthetic supervision should not stop at image-level visual plausibility. If one considered only Table 2, the Gaussian-only baseline would appear to be the strongest generator. The severity-controlled probe-side results show that this conclusion would be methodologically wrong.

Table 2: Unmatched generator-side ablation under canonical parameters. Higher PSNR/SSIM may reflect milder corruption rather than better supervision.

Condition	PSNR	SSIM	EdgeF1	MPI
A1 Full cascade	17.96	0.283	0.464	140.6
A2 P+G	19.73	0.361	0.507	144.0
A3 G-only	22.56	0.484	0.616	140.9
A4 S+G	18.81	0.341	0.512	133.1

5.3 Severity-Controlled Probe-Side Ablation

The key question is whether simpler baselines still provide better supervision once severity is controlled. Table 3 reports the severity-matched probe-side results under the expanded specimen-held-out protocol ($\sigma_{A2} = 0.186$, $\sigma_{A3} = 0.222$), specifically for the original seed-42 training-time pilot validation path. In these experiments the fixed probe, tile size, split definition, and training recipe are held constant, so the dominant manipulated variable is the corruption model itself. Only A2 and A3 are included in the matched probe study, because the matched analysis is designed to test whether removing the speckle and/or Poisson stages can be compensated by increasing Gaussian severity; A4 is therefore retained only in the unmatched generator-side comparison. A4 was excluded from the matched probe study because, under the present matching protocol, its retained speckle stage makes a Gaussian-only severity adjustment less informative than for A2 and A3; its main ablation role is therefore captured in the unmatched generator-side comparison.

The full cascade achieves the best validation loss and PSNR in the original seed-42 pilot, while matched simpler baselines retain small advantages on SSIM and EdgeF1. This pattern was already directionally consistent with the original 240-pair random split (matched losses: 0.0868, 0.0896, 0.0926), providing initial evidence that the finding was not a dataset-specific artifact. The matching procedure itself behaved as intended. On the training-side search target, the final matched settings reached 18.3009 dB and 18.3029 dB versus the A1 target of 18.2981 dB, with residual SSIM gaps of approximately 0.029 and 0.034. On the validation-side preflight, the matched baselines remained reasonably close to the A1 target (18.07 and 18.10 dB versus 17.96 dB), confirming that the probe-side comparison is cleanly PSNR-matched rather than jointly matched on every statistic.

Qualitatively, the full-cascade supervision teaches the probe to suppress speckle-driven texture while preserving the broader fringe

envelope, whereas the matched Gaussian-only baseline retains slightly crisper local edges at the cost of residual low-frequency artifacts that the full-cascade probe resolves. The visual examples in Figure 2 are consistent with this reading. Under full-cascade supervision, the probe learns to suppress granular speckle texture while maintaining the global fringe envelope and the large-scale phase-dependent morphology. Under Gaussian-only supervision, some local contour transitions remain slightly sharper, which is reflected in EdgeF1, but the output also preserves low-frequency residue and less coherent broad-scale structure. The correct interpretation is therefore not that the full cascade dominates on every metric, but that it is more probe-aligned than the reduced baselines when the comparison is made under controlled severity.

We further strengthened the specimen-held-out pilot by extending the matched probe-side comparison to three seeds under the same frozen protocol (same manifests, same W01 held-out specimen, same tile size, same fixed V4 probe, and identical training recipe). Under the original training-time validation path, the full cascade achieved the best mean validation loss across seeds (0.1046 ± 0.0176 versus 0.1075 ± 0.0193 and 0.1101 ± 0.0204 for the matched Poisson+Gaussian and Gaussian-only baselines, respectively), but the A1-versus-A2 comparison was not fully direction-stable because the sign of loss(A2)-loss(A1) flipped for one seed. By contrast, A1 remained directionally favorable against A3 on the primary validation-loss criterion for all three seeds.

We then re-evaluated the resulting best checkpoints post hoc on the explicit held-out W01 noisy/clean pair manifests. Under this stricter pair-manifest protocol, the full cascade again achieved the best mean validation loss across seeds (0.2080 ± 0.0382 versus 0.2182 ± 0.0346 and 0.2221 ± 0.0362), and now yielded lower validation loss than both matched baselines for all examined seeds. In other words, the A1-versus-A2 comparison is seed-sensitive under the original training-time pilot validation path, but becomes directionally consistent in favor of the full cascade once held-out utility is measured explicitly on the real noisy/clean W01 pairs. This supplementary result does not replace the original pilot validation summaries, since checkpoint selection was still governed by the training-time protocol; rather, it strengthens the interpretation that the full corruption chain is better aligned with the primary probe objective under a stricter held-out pair-based check.

5.4 Carbon-Composite Material-Shift Validation

To test whether the favorable severity-controlled ranking remained visible under bounded material shift, we constructed a specimen-level carbon-composite validation package. After deterministic normalization of four filename naming inconsistencies, the audited carbon subset became fully pairable, yielding 714/714 averaged/single-shot pairs across three independent specimens. Carbon calibration then reused the same moment-matching logic as the frozen wood baseline on this audited subset and produced meaningfully different carbon-domain parameters ($k = 1.9908$, $p = 16.0772$, $\sigma = 0.0720$) relative to the wood reference ($k = 2.1112$, $p = 24.7652$, $\sigma = 0.0861$). We therefore treated carbon as a bounded material-shift target rather than as an in-domain extension or a test of material invariance.

Using the existing wood-trained fixed V4 DnCNN-Lite-ECA checkpoints for `A1_full`, `A2_poisson_gaussian_matched`, and `A3_gaussian_only_matched`, we then evaluated transfer to the carbon domain without any carbon fine-tuning, retraining, or architectural change. Transfer was evaluated for all three available wood seeds (42, 13, and 37), and the carbon evaluation set remained fixed across seeds and conditions. On the primary metric, `A1_full` remained directionally favorable relative to both matched reduced baselines in all three seeds (Table 5). The mean validation loss across seeds was 0.210688 ± 0.019143 for A1 (full cascade), compared with 0.215312 ± 0.012311 for A2 (matched Poisson+Gaussian) and 0.217392 ± 0.012455 for A3 (matched Gaussian-only). At specimen level, the direction was fully stable against A3 and mostly stable against A2, with one small exception for specimen `C02` at seed 13. This aggregate transfer pattern was further reinforced by post hoc paired image-level analysis on the audited carbon evaluation set: for the primary validation-loss criterion, the paired loss deltas remained favorable to A1 against both reduced baselines in all three seeds, with strictly positive bootstrap confidence intervals and Wilcoxon signed-rank p-values below 0.01 throughout. A plausible interpretation is that the full cascade preserves a nuisance composition that is less tightly coupled to material-specific surface texture than the reduced baselines. Secondary metrics remained metric-dependent, so the carbon-composite package provides bounded transfer support for the full calibrated corruption regime on the primary probe-side objective rather than a claim of material invariance or universal superiority across all metrics.

5.5 Reproducibility-Oriented v3.2 Validation

The v3.2 branch was introduced as a methodology upgrade rather than a new generator. Its three core properties – backward compatibility, order invariance, and replayability – were all confirmed by 30/30 byte-identical outputs with maximum deviation 0.0. A regime-aware calibration mode was also added, yielding distinct parameter blocks for Mid ($k = 2.015$, $p = 15.221$, $\sigma = 0.084$) and High ($k = 2.224$, $p = 27.184$, $\sigma = 0.087$) while preserving the frozen baseline path. Together, these checks show that reproducibility and regime-awareness can coexist without altering the historical baseline.

At the same time, the practical utility signal of v3.2 remained modest: generator-side shifts were small (ΔPSNR = -0.042, ΔSSIM = -0.0022, ΔEdgeF1 = -0.0030), and probe-side changes were similarly limited (ΔPSNR = -0.036 dB, ΔSSIM = -0.0024, ΔEdgeF1 = -0.0015). The value of v3.2 is therefore methodological traceability rather than a large average quality gain. In that sense, v3.2 should be read as an experimental-infrastructure contribution: it makes the generator auditable, deterministic, and regime-aware without rewriting the scientific baseline.

5.6 Cross-Scale and Cross-Material Interpretation of the Ablation Claim

Taken together, the random-split study and the specimen-held-out pilot support a disciplined reading of the ablation claim. The paper does not argue that the full cascade is universally superior, nor that the reduced baselines are uninformative. Rather, it shows that once a major severity confound is removed, the full cascade remains

Table 3: Severity-controlled probe-side ablation under the expanded specimen-held-out protocol, reported for the original seed-42 training-time pilot validation path (train: W02+W03, 456 images; val: W01, 243 images; tile size 256; fixed V4 probe). Bold indicates the best value per column.

Condition	Best val. loss	Val. PSNR (dB)	Val. SSIM	Val. EdgeF1
A1 Full cascade	**0.0850**	**26.39**	0.766	0.740
A2 P+G matched ($\sigma = 0.186$)	0.0853	26.00	**0.770**	0.761
A3 G-only matched ($\sigma = 0.222$)	0.0866	25.59	**0.770**	**0.765**

Table 4: Protocol-separated multi-seed validation-loss deltas for the A1-versus-A2 specimen-held-out comparison. Positive Δ favors the full cascade on the primary metric.

Protocol	Seed 42	Seed 13	Seed 37	Mean ± std
Training-time pilot validation	+0.000247	+0.010409	−0.001906	+0.002917 ± 0.006577
Explicit pair-manifest reevaluation	+0.010383	+0.003397	+0.016946	+0.010242 ± 0.006776

Table 5: Bounded wood-to-carbon transfer evaluation using wood-trained fixed-probe checkpoints and an evaluation-only carbon set of 714 audited noisy/averaged pairs across three independent specimens. Values are mean ± standard deviation across seeds 42, 13, and 37; the aggregate ranking is further supported by paired image-level analysis reported in the text.

Condition	Mean val. loss ↓
A1 Full cascade	0.210688 ± 0.019143
A2 P+G matched	0.215312 ± 0.012311
A3 G-only matched	0.217392 ± 0.012455

better aligned with the fixed probe's primary optimization criterion at two independent scales. This matters because the two scales stress different failure modes: the smaller random split is more vulnerable to validation noise, whereas the held-out pilot is more sensitive to specimen-level mismatch.

At the same time, the consistency is qualitative rather than exact. The absolute losses differ across the two protocols, the validation populations are not interchangeable, and the held-out pilot introduces a stronger domain boundary. For this reason, the cross-scale agreement should be interpreted as converging evidence rather than as formal statistical confirmation. Its value lies in reducing the plausibility of a narrow artifact explanation. If the full cascade had only led under one split definition, the result could have been dismissed as a protocol accident; the fact that the same direction reappears under a stricter held-out setting makes that interpretation less convincing.

This cross-scale view also clarifies the metric-dependent nature of the result. The reduced baselines still recover slightly higher SSIM or EdgeF1 under some matched conditions, which means that they retain local structural advantages in specific respects. Yet the full cascade's consistent lead on the probe's primary objective suggests that matching the composition of nuisance processes can matter more for supervision quality than maximizing any single image-level similarity score. The multi-seed extension sharpens this reading: the full cascade remains directionally stable against the Gaussian-only matched reduction across both evaluation protocols, whereas its comparison against the Poisson+Gaussian matched reduction is seed-sensitive under the original training-time path but becomes consistently favorable under explicit held-out pair-manifest reevaluation. In practical terms, the lesson is not "always use the most complex corruption model," but rather "test whether added corruption structure survives severity control and fixed-probe evaluation."

6 Discussion

The main lesson is that synthetic supervision in ESPI should be treated as a modeling problem rather than as a trivial data-augmentation step. The unmatched ablation shows that image-level similarity alone is a poor guide: weaker corruptions can appear better because they are less severe. Once the comparison is controlled for severity, the full cascade becomes better aligned with the fixed probe's primary objective, suggesting that corruption *structure* – not only corruption magnitude – matters for useful supervision. The physics-informed structure of the cascade plays a specific role in this result. In our generator-side decomposition, the multiplicative speckle stage accounts for approximately 58% of the total corruption intensity, whereas the Gaussian component alone contributes less than 9%. This indicates that the full cascade does not merely produce "harder" corruptions; it produces corruptions whose statistical composition more closely reflects the real acquisition process. The probe-side advantage of the full cascade under severity control is therefore consistent with the interpretation that supervision quality depends not only on matching aggregate intensity, but also on matching the composition of physical nuisance sources.

At the same time, the paper makes bounded claims. The current evidence is specimen-family-specific, the High-regime result is asymmetric, and the full cascade is not uniformly superior across all metrics. These boundaries define the scope within which the proposed framework should be interpreted.

More broadly, the paper argues that synthetic supervision should be evaluated as an experimental object in its own right. In domains where paired real data are scarce, progress can come not only from larger denoisers or stronger loss functions but also from better-designed corruption models, tighter evaluation protocols, and clearer distinctions between plausibility, severity, and utility. The strengthened pilot further suggests that in synthetic-supervision studies the validation design itself should be treated as part of the experimental object rather than as a neutral implementation detail. ESPI is a demanding testbed for this question, but the

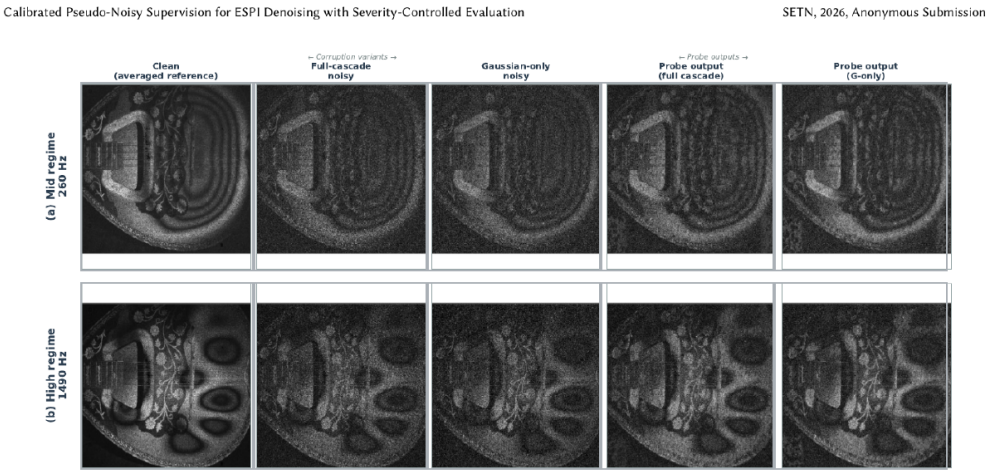


Figure 2: Representative held-out validation examples. Top row: Mid regime (260 Hz); bottom row: High regime (1490 Hz). Columns show the averaged reference, the full-cascade noisy input, the severity-matched Gaussian-only noisy input, and the fixed V4 probe outputs. The underlying averaged references are drawn from the ESPI acquisition framework of Bakarezos et al. [21], while the noisy and denoised panels are generated in the present work.

same methodological principle is relevant to other scientific imaging settings with structured noise and limited paired supervision. The practical lesson is not that one should always prefer the most elaborate corruption model, but that one should only prefer it once its advantage survives severity control and fixed-probe evaluation. Under bounded wood-to-carbon material shift, the full calibrated corruption regime also retained the most favorable direction on the primary probe-side objective, and paired image-level carbon analysis reinforced this transfer-based reading, suggesting that its utility is not confined to the original wood-board setting. A related lesson concerns experimental governance. Once synthetic supervision is treated as a first-class scientific object, provenance and replayability stop being mere engineering conveniences. They become part of the validity claim. The v3.2 layer is valuable precisely because it reduces the risk that hidden execution-order effects or silent routing differences contaminate the interpretation of a corruption-model comparison. For AI applications in scientific imaging, where data are scarce and regeneration is common, this is a non-trivial methodological gain.

7 Limitations and Future Work

The present study makes deliberately bounded claims. First, the calibration and strengthened pilot are specimen-family-specific: the available real single-shot / averaged pairs come from a restricted acquisition context and do not yet support a claim of broad cross-domain transfer. Second, the severity-controlled ablation is now strengthened across three seeds but still within a single specimen-held-out fold. The cross-scale agreement between the 240-pair random split and the 699-image held-out pilot is encouraging, but it should not be mistaken for formal statistical confirmation. Additional held-out folds and broader acquisition contexts would further strengthen the empirical basis of the ranking.

Third, the High regime result remains asymmetric. Although the generator stays learnable in that operating region, the High comparison uses a different tile size and a smaller validation set, and the diagnostic evidence points to a change in clean-reference structure as well as corruption profile. For this reason, the High result should be read as bounded evidence of regime extension rather than as a clean head-to-head benchmark against Mid. Fourth, the fixed-probe methodology is a strength for causal interpretation, but it also intentionally narrows the question being asked. The present paper does not claim that the same corruption ranking must hold for every denoising backbone or loss function; rather, it shows that the ranking is visible and methodologically meaningful for a stable lightweight probe.

The strengthened pilot analysis still has bounded scope. The multi-seed extension was carried out within the same specimen-held-out pilot and therefore does not yet constitute a full multi-fold confirmation campaign. In addition, the stricter pair-manifest reevaluation was performed post hoc on checkpoints selected under the original training-time synthetic validation path; it should therefore be interpreted as a complementary held-out utility check rather than as a direct numerical replacement of the training-time validation summaries. The carbon-composite extension likewise remains an evaluation-only transfer study, so it should be interpreted as bounded material-shift evidence rather than as a claim of material invariance or uniformly superior generalization, especially given the single `C02`/seed-13 exception against `A2`.

These limitations suggest a clear next research program. The first priority is statistical stabilization of the severity-controlled finding through additional seeds and held-out folds. The second is broader validation across acquisition families and specimen types to determine how far the calibrated pseudo-noisy framework transfers before re-calibration becomes necessary. A third direction is to study whether the fixed-probe paradigm can be generalized beyond ESPI to other scientific-imaging domains where structured noise,

scarce paired data, and synthetic-to-real mismatch co-occur. The broader methodological question is not limited to interferometry: it concerns how synthetic supervision should be trusted, audited, and compared when it becomes a major source of training signal.

8 Conclusion

We presented a calibrated pseudo-noisy supervision framework for ESPI denoising based on a physics-informed corruption cascade and fixed-probe evaluation. Severity-controlled evaluation proved necessary to avoid misleading conclusions from generator-side fidelity alone: the full corruption chain is more probe-aligned than matched simpler baselines, although the advantage remains metric-dependent. A bounded carbon-composite transfer study further suggested that this favorable ranking is not confined to the original wood-board setting, and paired image-level carbon analysis reinforced this transfer-based reading, while still falling short of material invariance. Beyond the specific ESPI setting, the paper argues for a broader methodological principle: under domain mismatch, synthetic supervision should be calibrated, evaluated functionally, and made reproducible by design. A natural next step is to extend the strengthened multi-seed analysis to additional held-out folds and acquisition families, and to unify training-time synthetic summaries with explicit held-out pair-manifest metrics within the same controlled validation design.

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